

T3 BSM UV terms grouped by field content

EFT order: 5; loop order: 1. Columns are determined only from BSM UV terms. Empty cells denote no UV term with that field combination.

Field combinations 1 of 3

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	S_1	S_2	$F D$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{1}{2} N_{r_1}^T C N_{r_1} M_F$	$-\text{NewScalar1} \text{NewScalar1}_{a_1} M_F^{\dagger}$ $+\frac{1}{2} \text{Imbhd} \text{d} \text{Inv1} \text{NewScalar1}_{a_1} \text{NewScalar1}_{a_1} M_F^{\dagger}$	$-\text{NewScalar1} \text{NewScalar1}_{a_1} M_F^{\dagger}$ $+\frac{1}{2} \text{Imbhd} \text{d} \text{Inv1} \text{NewScalar1}_{a_1} \text{NewScalar1}_{a_1} M_F^{\dagger}$	$+\frac{1}{2} i N_{r_1}^T C \gamma^{\mu_1} D_{\mu_1} (N_{r_1})$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{1}{2} N_{A_1, r_1}^T C N_{A_1, r_1} M_F$	$-\text{NewScalar1} \text{NewScalar1}_{a_1} M_F^{\dagger}$ $+\frac{1}{2} \text{Imbhd} \text{d} \text{Inv1} \text{NewScalar1}_{a_1} \text{NewScalar1}_{a_1} M_F^{\dagger}$	$-\text{NewScalar1} \text{NewScalar1}_{a_1} M_F^{\dagger}$ $+\frac{1}{2} \text{Imbhd} \text{d} \text{Inv1} \text{NewScalar1}_{a_1} \text{NewScalar1}_{a_1} M_F^{\dagger}$	$+\frac{1}{2} i N_{A_1, r_1}^T C \gamma^{\mu_1} D_{\mu_1} (N_{A_1, r_1})$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\overline{N_{a_1, r_1}} N_{a_1, r_1} M_F$	$-\text{NewScalar1} \text{NewScalar1}_{a_1} M_F^{\dagger}$ $+\frac{1}{2} \text{Imbhd} \text{d} \text{Inv1} \text{NewScalar1}_{a_1} \text{NewScalar1}_{a_1} M_F^{\dagger}$	$-\text{NewScalar1} \text{NewScalar1}_{a_1} M_F^{\dagger}$ $+\frac{1}{2} \text{Imbhd} \text{d} \text{Inv1} \text{NewScalar1}_{a_1} \text{NewScalar1}_{a_1} M_F^{\dagger}$	$+i \overline{N_{a_1, r_1}} \gamma^{\mu_1} D_{\mu_1} (N_{a_1, r_1})$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\overline{N_{a_1, r_1}} N_{a_1, r_1} M_F$	$-\text{NewScalar1} \text{NewScalar1}_{a_1} M_F^{\dagger}$ $+\frac{1}{2} \text{Imbhd} \text{d} \text{Inv1} \text{NewScalar1}_{a_1} \text{NewScalar1}_{a_1} M_F^{\dagger}$	$-\text{NewScalar1} \text{NewScalar1}_{a_1} M_F^{\dagger}$ $+\frac{1}{2} \text{Imbhd} \text{d} \text{Inv1} \text{NewScalar1}_{a_1} \text{NewScalar1}_{a_1} M_F^{\dagger}$	$+i \overline{N_{a_1, r_1}} \gamma^{\mu_1} D_{\mu_1} (N_{a_1, r_1})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\overline{N_{a_1, r_1}} N_{a_1, r_1} M_F$	$-\text{NewScalar1} \text{NewScalar1}_{a_1} M_F^{\dagger}$ $+\frac{1}{2} \text{Imbhd} \text{d} \text{Inv1} \text{NewScalar1}_{a_1} \text{NewScalar1}_{a_1} M_F^{\dagger}$	$-\text{NewScalar1} \text{NewScalar1}_{a_1} M_F^{\dagger}$ $+\frac{1}{2} \text{Imbhd} \text{d} \text{Inv1} \text{NewScalar1}_{a_1} \text{NewScalar1}_{a_1} M_F^{\dagger}$	$+i \overline{N_{a_1, r_1}} \gamma^{\mu_1} D_{\mu_1} (N_{a_1, r_1})$

Field combinations 2 of 3

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	D	S_1	H	S_1	S_2	S_2	D
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+D_{\mu_1}$	$(\text{NewScalar1}_{a_1})^\dagger D_{\mu_1} (\text{NewScalar1}_{a_1})$	$+\text{im bdaH1Inv1}H_{a_2}\text{NewScalar1}_{a_4}\text{T3HiggsPortalICGInv1}_{a_1,a_2,a_2,a_4}H_{a_1}^{\dagger}\text{NewScalar1}_{a_2}^{\dagger}$ $+\text{im bdaH1Inv2}H_{a_2}\text{NewScalar1}_{a_4}\text{T3HiggsPortalICGInv2}_{a_1,a_2,a_2,a_4}H_{a_1}^{\dagger}\text{NewScalar1}_{a_2}^{\dagger}$	$+\text{im bda12Inv1NewScalar1}_{a_4}\text{w5color1}_{a_4}\text{T1CrossScalarCGInv1}_{a_1,a_2,a_2,a_4}\text{w5color1}_{a_4}^{\dagger}\text{NewScalar1}_{a_2}^{\dagger}$ $+\text{im bda12Inv2NewScalar1}_{a_4}\text{w5color1}_{a_4}\text{T1CrossScalarCGInv2}_{a_1,a_2,a_2,a_4}\text{w5color1}_{a_4}^{\dagger}\text{NewScalar1}_{a_2}^{\dagger}$	$+D_{\mu_1} (\text{NewScalar2}_{a_1})^\dagger D_{\mu_1} (\text{NewScalar2}_{a_1})$			
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+D_{\mu_1}$	$(\text{NewScalar1}_{a_1})^\dagger D_{\mu_1} (\text{NewScalar1}_{a_1})$	$+\text{im bdaH1Inv1}H_{a_2}\text{NewScalar1}_{a_4}\text{T3HiggsPortalICGInv1}_{a_1,a_2,a_2,a_4}H_{a_1}^{\dagger}\text{NewScalar1}_{a_2}^{\dagger}$ $+\text{im bdaH1Inv2}H_{a_2}\text{NewScalar1}_{a_4}\text{T3HiggsPortalICGInv2}_{a_1,a_2,a_2,a_4}H_{a_1}^{\dagger}\text{NewScalar1}_{a_2}^{\dagger}$	$+\text{im bda12Inv1NewScalar1}_{a_4}\text{w5color1}_{a_4}\text{T1CrossScalarCGInv1}_{a_1,a_2,a_2,a_4}\text{w5color1}_{a_4}^{\dagger}\text{NewScalar1}_{a_2}^{\dagger}$ $+\text{im bda12Inv2NewScalar1}_{a_4}\text{w5color1}_{a_4}\text{T1CrossScalarCGInv2}_{a_1,a_2,a_2,a_4}\text{w5color1}_{a_4}^{\dagger}\text{NewScalar1}_{a_2}^{\dagger}$	$+D_{\mu_1} (\text{NewScalar2}_{a_1})^\dagger D_{\mu_1} (\text{NewScalar2}_{a_1})$			
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1}$	$(\text{NewScalar1})^\dagger D_{\mu_1} (\text{NewScalar1})$	$+\text{im bdaH1Inv1NewScalar1}H_{a_2}\text{T3HiggsPortalICGInv1}_{a_1,a_2}\text{NewScalar1}^{\dagger}H_{a_1}^{\dagger}$	$+\text{im bda12Inv1NewScalar1New8color1}_{a_4}\text{T3CrossScalarCGInv1}_{a_1,a_2,a_2,a_4}\text{NewScalar1}^{\dagger}\text{New8color1}_{a_1}^{\dagger}$	$+D_{\mu_1} (\text{NewScalar2}_{A_1})^\dagger D_{\mu_1} (\text{NewScalar2}_{A_1})$			
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+D_{\mu_1}$	$(\text{NewScalar1}_{A_1})^\dagger D_{\mu_1} (\text{NewScalar1}_{A_1})$	$+\frac{1}{2}\text{im bdaH1Inv1}H_{a_2}\text{NewScalar1}_{A_2}\text{T3HiggsPortalICGInv1}_{a_1,a_2,A_1,A_2}H_{a_1}^{\dagger}\text{NewScalar1}_{A_1}^{\dagger}$ $+\frac{1}{2}\text{im bdaH1Inv2}H_{a_2}\text{NewScalar1}_{A_2}\text{T3HiggsPortalICGInv2}_{a_1,a_2,A_1,A_2}H_{a_1}^{\dagger}\text{NewScalar1}_{A_1}^{\dagger}$	$+\text{im bda12Inv1NewScalar1New8color1}_{A_2}\text{T3CrossScalarCGInv1}_{a_1,a_2,a_2,a_4}\text{NewScalar1}^{\dagger}\text{New8color1}_{A_1}^{\dagger}$	$+D_{\mu_1} (\text{NewScalar2})^\dagger D_{\mu_1} (\text{NewScalar2})$			
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1}$	$(\text{NewScalar1}_{A_1})^\dagger D_{\mu_1} (\text{NewScalar1}_{A_1})$	$+\frac{1}{2}\text{im bdaH1Inv1}H_{a_2}\text{NewScalar1}_{A_2}\text{T3HiggsPortalICGInv1}_{a_1,a_2,A_1,A_2}H_{a_1}^{\dagger}\text{NewScalar1}_{A_1}^{\dagger}$ $+\frac{1}{2}\text{im bdaH1Inv2}H_{a_2}\text{NewScalar1}_{A_2}\text{T3HiggsPortalICGInv2}_{a_1,a_2,A_1,A_2}H_{a_1}^{\dagger}\text{NewScalar1}_{A_1}^{\dagger}$ $+\frac{1}{2}\text{im bda12Inv1NewScalar1}_{a_4}\text{w5color1}_{a_4}\text{T1CrossScalarCGInv1}_{a_1,a_2,a_2,a_4}\text{w5color1}_{a_4}^{\dagger}\text{NewScalar1}_{a_2}^{\dagger}$ $+\frac{1}{2}\text{im bda12Inv2NewScalar1}_{a_4}\text{w5color1}_{a_4}\text{T1CrossScalarCGInv2}_{a_1,a_2,a_2,a_4}\text{w5color1}_{a_4}^{\dagger}\text{NewScalar1}_{a_2}^{\dagger}$	$+D_{\mu_1} (\text{NewScalar2}_{A_1})^\dagger D_{\mu_1} (\text{NewScalar2}_{A_1})$				

Field combinations 3 of 3

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_2 H$	$S_1 F \ell$	$S_1 S_2 H$	$S_2 F \ell$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+i\text{ambd}H_2\text{Inv}1H_{a_2}\text{NewScalar}2_{a_4}\text{T3HiggsPortal}2\text{CGInv}1_{a_1,a_2,a_3,a_4}H_{a_1}{}^\dagger\text{NewScalar}2_{a_2}{}^\dagger$ $+i\text{ambd}H_2\text{Inv}2H_{a_2}\text{NewScalar}2_{a_4}\text{T3HiggsPortal}2\text{CGInv}2_{a_1,a_2,a_3,a_4}H_{a_1}{}^\dagger\text{NewScalar}2_{a_2}{}^\dagger$	$+\overline{\ell_{a_1,i_1}}P_RN_{r_1}\text{NewScalar}1_{a_1}y_{1,i_1,r_1}$ $+\overline{y_{1,i_1,r_1}}N_{r_1}{}^TC_P L\ell_{a_1,i_1}\text{NewScalar}1_{a_1}{}^\dagger$	$+\overline{\lambda_{T3}}\text{NewScalar}2_{a_1}\epsilon_{a_2,a_3}H_{a_1}{}^\dagger H_{a_2}{}^\dagger\text{NewScalar}1_{a_3}{}^\dagger$ $+H_{a_1}H_{a_2}\text{NewScalar}1_{a_3}\lambda_{T3}\bar{\epsilon}_{a_2,a_3}\text{NewScalar}2_{a_1}{}^\dagger$	$+\overline{y_{2,i_1,r_1}}N_{r_1}{}^TC_P L\ell_{a_1,i_1}\text{NewScalar}2_{a_2}\bar{\epsilon}_{a_1,a_2}$ $+\overline{\ell_{a_2,i_1}}P_RN_{r_1}y_{2,i_1,r_1}\epsilon_{a_2,a_1}\text{NewScalar}2_{a_1}{}^\dagger$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+i\text{ambd}H_2\text{Inv}1H_{a_2}\text{NewScalar}2_{a_4}\text{T3HiggsPortal}2\text{CGInv}1_{a_1,a_2,a_3,a_4}H_{a_1}{}^\dagger\text{NewScalar}2_{a_2}{}^\dagger$ $+i\text{ambd}H_2\text{Inv}2H_{a_2}\text{NewScalar}2_{a_4}\text{T3HiggsPortal}2\text{CGInv}2_{a_1,a_2,a_3,a_4}H_{a_1}{}^\dagger\text{NewScalar}2_{a_2}{}^\dagger$	$+\overline{\ell_{a_1,i_1}}P_RN_{A_1,r_1}\text{NewScalar}1_{a_2}y_{1,i_1,r_1}\text{gen}(\text{SU}2L(\text{fund}))_{A_1,a_1,a_2}$ $+\overline{y_{1,i_1,r_1}}N_{A_1,r_1}{}^TC_P L\ell_{a_2,i_1}\text{gen}(\text{SU}2L(\text{fund}))_{A_1,a_1,a_2}\text{NewScalar}1_{a_1}{}^\dagger$	$+\overline{\lambda_{T3}}\text{NewScalar}2_{a_1}\epsilon_{a_2,a_3}H_{a_1}{}^\dagger H_{a_2}{}^\dagger\text{NewScalar}1_{a_3}{}^\dagger$ $+H_{a_1}H_{a_2}\text{NewScalar}1_{a_3}\lambda_{T3}\bar{\epsilon}_{a_2,a_3}\text{NewScalar}2_{a_1}{}^\dagger$	$+\overline{y_{2,i_1,r_1}}N_{A_1,r_1}{}^TC_P L\ell_{a_3,i_1}\text{NewScalar}2_{a_2}\text{gen}(\text{SU}2L(\text{fund}))_{A_1,a_1,a_3}\bar{\epsilon}_{a_1,a_2}$ $+\overline{\ell_{a_3,i_1}}P_RN_{A_1,r_1}y_{2,i_1,r_1}\epsilon_{a_2,a_1}\text{gen}(\text{SU}2L(\text{fund}))_{A_1,a_3,a_2}\text{NewScalar}2_{a_1}{}^\dagger$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{2}i\text{ambd}H_2\text{Inv}1H_{a_2}\text{NewScalar}2_{a_4}\text{T3HiggsPortal}2\text{CGInv}1_{a_1,a_2,a_3,a_4}H_{a_1}{}^\dagger\text{NewScalar}2_{a_1}{}^\dagger$ $+\frac{1}{2}i\text{ambd}H_2\text{Inv}1H_{a_2}\text{NewScalar}2_{a_4}\text{T3HiggsPortal}2\text{CGInv}1_{a_1,a_2,a_3,a_4}H_{a_1}{}^\dagger\text{NewScalar}2_{a_1}{}^\dagger$ $+\frac{1}{2}i\text{ambd}H_2\text{Inv}2H_{a_2}\text{NewScalar}2_{a_4}\text{T3HiggsPortal}2\text{CGInv}2_{a_1,a_2,a_3,a_4}H_{a_1}{}^\dagger\text{NewScalar}2_{a_1}{}^\dagger$ $+\frac{1}{2}i\text{ambd}H_2\text{Inv}2H_{a_2}\text{NewScalar}2_{a_4}\text{T3HiggsPortal}2\text{CGInv}2_{a_1,a_2,a_3,a_4}H_{a_1}{}^\dagger\text{NewScalar}2_{a_1}{}^\dagger$	$+\text{NewScalar}1\overline{\ell_{a_1,i_1}}C_P RN_{a_2,r_1}{}^Ty_{1,i_1,r_1}\epsilon_{a_1,a_2}$ $+\overline{y_{1,i_1,r_1}}\ell_{a_1,i_1}{}^TC_P LN_{a_2,r_1}\bar{\epsilon}_{a_1,a_2}\text{NewScalar}1{}^\dagger$	$+\overline{\lambda_{T3}}\text{NewScalar}2_{A_1}\epsilon_{a_3,a_2}\text{gen}(\text{SU}2L(\text{fund}))_{A_1,a_1,a_3}\text{NewScalar}1{}^\dagger H_{a_1}{}^\dagger H_{a_2}{}^\dagger$ $+\text{NewScalar}1H_{a_2}H_{a_3}\lambda_{T3}\text{gen}(\text{SU}2L(\text{fund}))_{A_1,a_1,a_3}\bar{\epsilon}_{a_1,a_2}\text{NewScalar}2_{A_1}{}^\dagger$	$+\overline{y_{2,i_1,r_1}}\overline{N_{a_1,r_1}}P_L\ell_{a_2,i_1}\text{NewScalar}2_{A_1}\text{gen}(\text{SU}2L(\text{fund}))_{A_1,a_1,a_2}$ $+\overline{\ell_{a_1,i_1}}P_RN_{a_2,r_1}y_{2,i_1,r_1}\text{gen}(\text{SU}2L(\text{fund}))_{A_1,a_1,a_2}\text{NewScalar}2_{A_1}{}^\dagger$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+i\text{ambd}H_2\text{Inv}1\text{NewScalar}2H_{a_2}\text{T3HiggsPortal}2\text{CGInv}1_{a_1,a_2}\text{NewScalar}2{}^\dagger H_{a_1}{}^\dagger$	$+\overline{\ell_{a_3,i_1}}C_P RN_{a_1,r_1}{}^T\text{NewScalar}1_{A_1}y_{1,i_1,r_1}\epsilon_{a_1,a_2}\text{gen}(\text{SU}2L(\text{fund}))_{A_1,a_3,a_2}$ $+\overline{y_{1,i_1,r_1}}\ell_{a_3,i_1}{}^TC_P LN_{a_1,r_1}\text{gen}(\text{SU}2L(\text{fund}))_{A_1,a_2,a_3}\bar{\epsilon}_{a_1,a_2}\text{NewScalar}1_{A_1}{}^\dagger$	$+H_{a_2}H_{a_3}\text{NewScalar}1_{A_1}\lambda_{T3}\text{gen}(\text{SU}2L(\text{fund}))_{A_1,a_1,a_3}\bar{\epsilon}_{a_1,a_2}\text{NewScalar}2{}^\dagger$ $+\text{NewScalar}2\overline{\lambda_{T3}}\epsilon_{a_3,a_2}\text{gen}(\text{SU}2L(\text{fund}))_{A_1,a_1,a_3}H_{a_1}{}^\dagger H_{a_2}{}^\dagger\text{NewScalar}1_{A_1}{}^\dagger$	$+\text{NewScalar}2y_{2,i_1,r_1}\overline{N_{a_1,r_1}}P_L\ell_{a_1,i_1}$ $+\overline{\ell_{a_1,i_1}}P_RN_{a_1,r_1}y_{2,i_1,r_1}\text{NewScalar}2{}^\dagger$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{2}i\text{ambd}H_2\text{Inv}1H_{a_2}\text{NewScalar}2_{a_4}\text{T3HiggsPortal}2\text{CGInv}1_{a_1,a_2,a_3,a_4}H_{a_1}{}^\dagger\text{NewScalar}2_{A_1}{}^\dagger$ $+\frac{1}{2}i\text{ambd}H_2\text{Inv}1H_{a_2}\text{NewScalar}2_{a_4}\text{T3HiggsPortal}2\text{CGInv}1_{a_1,a_2,a_3,a_4}H_{a_1}{}^\dagger\text{NewScalar}2_{A_1}{}^\dagger$ $+\frac{1}{2}i\text{ambd}H_2\text{Inv}2H_{a_2}\text{NewScalar}2_{a_4}\text{T3HiggsPortal}2\text{CGInv}2_{a_1,a_2,a_3,a_4}H_{a_1}{}^\dagger\text{NewScalar}2_{A_1}{}^\dagger$ $+\frac{1}{2}i\text{ambd}H_2\text{Inv}2H_{a_2}\text{NewScalar}2_{a_4}\text{T3HiggsPortal}2\text{CGInv}2_{a_1,a_2,a_3,a_4}H_{a_1}{}^\dagger\text{NewScalar}2_{A_1}{}^\dagger$	$+\overline{\ell_{a_3,i_1}}C_P RN_{a_1,r_1}{}^T\text{NewScalar}1_{A_1}y_{1,i_1,r_1}\epsilon_{a_1,a_2}\text{gen}(\text{SU}2L(\text{fund}))_{A_1,a_3,a_2}$ $+\overline{y_{1,i_1,r_1}}\ell_{a_3,i_1}{}^TC_P LN_{a_1,r_1}\text{gen}(\text{SU}2L(\text{fund}))_{A_1,a_2,a_3}\bar{\epsilon}_{a_1,a_2}\text{NewScalar}1_{A_1}{}^\dagger$	$+\overline{\lambda_{T3}}\text{NewScalar}2_{a_2}\epsilon_{a_3,a_2}\text{gen}(\text{SU}2L(\text{fund}))_{A_1,a_1,a_3}H_{a_1}{}^\dagger\text{NewScalar}1_{A_1}{}^\dagger$ $-H_{a_2}H_{a_3}\text{NewScalar}1_{A_1}\lambda_{T3}\text{gen}(\text{SU}2L(\text{fund}))_{A_1,a_1,a_3}\bar{\epsilon}_{a_1,a_2}\text{NewScalar}2_{A_1}{}^\dagger$	$+\overline{y_{2,i_1,r_1}}\overline{N_{a_1,r_1}}P_L\ell_{a_2,i_1}\text{NewScalar}2_{A_1}\text{gen}(\text{SU}2L(\text{fund}))_{A_1,a_1,a_2}$ $+\overline{\ell_{a_1,i_1}}P_RN_{a_2,r_1}y_{2,i_1,r_1}\text{gen}(\text{SU}2L(\text{fund}))_{A_1,a_1,a_2}\text{NewScalar}2_{A_1}{}^\dagger$